Neel Gali

Software Engineer | Data Platforms

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Summary

Software engineer building **Python/SQL pipelines, Spark transformations, and production data products** across enterprise security, compliance SaaS, and scientific operations. Experience with Snowflake and Databricks, historical data modeling, schema evolution, data quality, and query optimization. Delivers tested, monitored datasets for analytics and AI workflows.

Technical Skills

Programming: Python, SQL, Spark SQL, Java, Bash
Data Platforms: Apache Spark/PySpark, Snowflake, Databricks, Delta Lake, PostgreSQL, Kafka
Pipelines & Quality: Apache Airflow, dbt, Great Expectations, Pandas, Pydantic, ETL/ELT, JSON, Parquet
Data Modeling: Dimensional Modeling, SCD Types 1/2, Star Schemas, Schema Evolution, Partitioning, Indexing
Engineering: AWS, Azure, Docker, Kubernetes, Terraform, Git, GitHub Actions, Jenkins, CI/CD, REST APIs

Professional Experience

Software Engineer AI *2025 - Present*
CIFC Asset Management | Enterprise Security Data & AI

- Built **Python, PySpark, and Airflow pipelines** ingesting security telemetry from 15+ systems into Snowflake, supplying curated data for security analytics and automated operational workflows.
- Developed **50+ modular dbt SQL models** with incremental materializations and data-quality tests, enabling self-service analytics and reducing ad hoc analyst queries by 60%.
- Modeled identity and access history with **SCD Type 2, effective dates, and surrogate keys** in PostgreSQL, preserving historical state for accurate compliance audit trails.
- Tuned **Snowflake transformations and clustering keys** and integrated Snowpipe ingestion, reducing downstream query latency by 45% for enterprise security reporting.
- Implemented **Great Expectations validation and pipeline alerting** to detect schema drift and anomalous data before propagation into production analytics datasets.
- Integrated curated security data into **FastAPI and AI incident-triage services**, processing 1,000+ daily events and reducing analyst workload by 30%.

Founding Engineer (Full Stack AI) *2023 - 2025*
Samay Systems LLC | B2B Compliance SaaS

- Delivered **Python/PySpark ETL pipelines** on AWS, ingesting compliance data from 10+ sources into Snowflake with lineage tracking for audit and reporting workflows.
- Designed **star schemas and SCD Type 1/2 models**, defining fact-table grain, conformed dimensions, and historical snapshots to represent changing compliance entities consistently.
- Built incremental **dbt transformations and 150+ data-quality tests**, documenting reusable SQL models for evidence collection across multi-tenant environments.
- Orchestrated **Airflow DAGs spanning 25+ tasks** with dependency management, retries, and SLA monitoring, achieving 99.5% success for critical nightly data loads.
- Implemented **Databricks/Delta Lake transformations** with idempotent writes, watermarking, and schema evolution; integrated Kafka event streams to decouple 6+ upstream services from ingestion.
- Optimized **PostgreSQL indexes, partitions, and materialized views** for compliance dashboards supporting 30K daily transactions, keeping application queries responsive as data volumes grew.

Software Engineer *2020 - 2022*
ZeMoSo Technologies | LabOps Data Platform

- Built **Python/Pandas and SQL ETL workflows** transforming experimental data into normalized PostgreSQL tables for a scientific platform used by 500+ researchers.
- Enforced **schema contracts and data-quality rules** with Pydantic validation, reducing downstream data errors by 40% in sample-tracking workflows.
- Designed **relational schemas, foreign-key constraints, and indexes** across 10+ modules for experiments, inventory, compounds, and batches, preserving traceability between scientific records.
- Automated testing and deployment through **GitHub Actions and Jenkins** for API services and application releases on AWS, reducing manual release time from 2 hours to 10 minutes.
- Worked with scientists and product managers to translate experimental workflows into **database schemas and API contracts**, aligning data capture and validation with regulated operational needs.

Education

Master's in Computer Science, Oklahoma State University *May 2024*
Bachelor's in Computer Science, JNTU Hyderabad *2021*